Project Portfolio: Mini-Redis Application Layer

5 Real-World Industry Applications Powered by a Custom C++ In-Memory Engine

Course: B.Tech 2nd Year CSE
Evaluator: Dr. Gunjan Chugh
Team Name: RediSetGo
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Executive Purpose: This document presents 5 distinct real-world software applications designed to run on top of our custom **Mini-Redis C++ in-memory database engine**. While standard database projects only perform basic terminal CRUD, these applications demonstrate practical utility in production systems (microsecond caching, session token authentication, traffic rate-limiting, and ephemeral storage) for college evaluation.

The Common Foundation: High-Performance C++17 Core Engine

Every application below connects directly over raw TCP sockets (Port 6379) to our C++ database engine.

`O(1) LRU Cache` `Doubly Linked List` `Thread Pool (POSIX/Win)` `AOF Durability` `0.018 ms Latency`

Application Architectures Portfolio

1 High-Performance URL Shortener & Analytics Platform (Bitly Style)

✓ FULLY BUILT & LIVE

Real-world analogue: Bitly, TinyURL — Sub-millisecond URL redirection with memory-bounded caching

Users input long URLs to generate concise short links (`/s/linux-kernel`). When requested, the service performs an in-memory lookup in 0.018ms, increments real-time click analytics, and executes an instant HTTP 302 redirect. Links support auto-expiry (TTL) and live QR code generation.

HOW IT USES MINI-REDIS C++ ENGINE:

- `SET url:<code> <dest> EX <ttl>` : Stores destination in RAM with optional auto-expiry.
- `GET url:<code>` : Constant-time O(1) resolution during HTTP 302 redirect.
- `INCR clicks:<code>` : Real-time atomic analytics counter per short link.
- `LRU Eviction` : Drops least-accessed links when memory capacity is reached.

LIVE DEMONSTRATION HIGHLIGHTS FOR PROFESSOR:

- **Instant 302 Redirect:** Browser redirects in < 1 ms from C++ RAM.
- **Real-Time Clicks Tracker:** Click counter increments live on dashboard.
- **Phone QR Scanner:** Evaluator can scan generated QR code on mobile camera.
- **TTL Auto-Purge:** 1-minute demo links disappear live as countdown reaches 0.

2 API Rate Limiter & Anti-Spam Security Shield (Cloudflare Style)

READY TO IMPLEMENT

Real-world analogue: Cloudflare, Stripe API Gateways — Traffic shaping and DDoS prevention

An API gateway middleware that protects web servers from brute-force attacks and request spamming. Each client IP address is restricted to a fixed quota (e.g., max 10 requests per minute). Requests exceeding the quota receive an immediate HTTP 429 Too Many Requests response.

HOW IT USES MINI-REDIS C++ ENGINE:

- `SET rate:<ip> 1 EX 60` : Initializes request counter with 60s sliding window.
- `INCR rate:<ip>` : Atomically tracks hit counts per incoming client request.
- `TTL rate:<ip>` : Checks remaining window time before counter resets.
- `Multi-threading` : Thread-safe mutexes handle high-concurrency attack spikes.

LIVE DEMONSTRATION HIGHLIGHTS FOR PROFESSOR:

- **Live Flooder Test:** Rapidly click a button 15 times in 5 seconds.
- **Security Lockout:** First 10 requests pass green; 11th triggers a red 429 Blocked card.
- **Auto-Unlock Countdown:** Live visual timer counts down until the client is unbanned.
- **Systems Value:** Directly demonstrates cybersecurity and traffic engineering.

3 Real-Time Multi-User Chat & Session Store (Discord / Slack Style)

READY TO IMPLEMENT

Real-world analogue: Discord, WhatsApp Web — Ephemeral messaging and authentication session tokens

A collaborative real-time messaging platform where user login sessions and recent conversation channels are held entirely in memory. If a user remains idle, their session token automatically expires. Message logs remain ephemeral in RAM without slow disk write overhead.

HOW IT USES MINI-REDIS C++ ENGINE:

- `SET session:<token> <user> EX 300` : Ephemeral authentication session token.
- `RPUSH channel:<id> <msg>` : Fast in-memory message history buffer.
- `Active TTL Sweep` : Background cleaner automatically purges inactive sessions.
- `TCP Sockets` : Socket connections handle concurrent chat participant streams.

LIVE DEMONSTRATION HIGHLIGHTS FOR PROFESSOR:

- **Multi-Window Chat:** Open two browser tabs side-by-side chatting in real time.
- **Session Invalidation:** Expire a session token to show security logout live.
- **Zero Disk Lag:** Demonstrates why chatting systems prefer in-memory engines over SQL.

4 Live Gaming Leaderboard & Real-Time Scoreboard (BGMI / Chess.com Style)

★ BUILT & LIVE

Real-world analogue: BGMI, Chess.com, HackerRank — Millisecond ranking updates under heavy concurrent load

A live competitive scoreboard that dynamically ranks active players based on scores. Traditional databases lock up when hundreds of players score simultaneously; Mini-Redis updates player rankings in memory and computes the top-10 leaderboard with zero perceptible latency.

HOW IT USES MINI-REDIS C++ ENGINE:

- `SET lb:score:<id> <score>` : Instant in-memory score tracking.
- `LRU Ranking Chain` : Visualizes top players ordered dynamically by score.
- `AOF Durability` : Prevents tournament data loss if game server restarts.

LIVE DEMONSTRATION HIGHLIGHTS FOR PROFESSOR:

- **Live Rank Shuffling:** Increment a player's score and watch their card slide up.
- **Simulated Match:** Click "Simulate Match" to inject 100 concurrent score updates.

5 Ephemeral OTP & 2-Factor Authentication Service (Zomato / Uber Style)

★ BUILT & LIVE

Real-world analogue: Zomato, Uber, Banking OTPs — Secure single-use verification tokens

Simulates an enterprise OTP service. When a user requests a login code, a 6-digit numeric token is saved in Mini-Redis with a strict 3-minute TTL. Once verified or expired, the key is permanently purged from memory so tokens cannot be intercepted or reused.

HOW IT USES MINI-REDIS C++ ENGINE:

- `SET otp:code:<phone> <pin> EX 45` : Auto-expiring one-time password in RAM.
- `DEL otp:code:<phone>` : Single-use invalidation upon successful validation.

LIVE DEMONSTRATION HIGHLIGHTS FOR PROFESSOR:

- **Login Flow:** Generate OTP → enter code → verify → verify again to show rejection.
- **Expired Token Test:** Wait for timer expiry and demonstrate unauthorized access block.

Architectural Comparison Matrix

APPLICATION IDEA	PRIMARY PROBLEM SOLVED	CORE C++ ENGINE FEATURES USED	VIVA / DEMO WOW FACTOR	CURRENT STATUS
1. URL Shortener (Bitly)	O(1) high-throughput link redirection	SET, GET, TTL expiry, LRU eviction, AOF logging	Instant 302 redirect, Live QR scan on phone	**★ Ready & Live**
2. API Rate Limiter	Anti-DDoS, brute-force spam protection	Atomic INCR, 60s sliding window TTL, Thread pool	Spam flood test, 429 Lockout visual timer	Concept Ready
3. Real-Time Chat & Sessions	Low-latency ephemeral messaging & auth	Session token TTL, in-memory buffer, TCP sockets	Dual-window chat, Session auto-expiry logout	Concept Ready
4. Gaming Leaderboard	High-frequency concurrent rank updates	Fast key updates, LRU access chains, AOF recovery	Live animated rank shuffling on score update	**★ Ready & Live**
5. Ephemeral OTP Service	Single-use temporary security tokens	Strict TTL auto-purge, Single-use DEL purge	OTP expiry rejection, Token replay rejection	**★ Ready & Live**

Flagship Viva Recommendation for Dr. Gunjan Chugh: We strongly recommend evaluating **AuthKV (Ephemeral 2FA & Identity Security Gateway)**. Backed by our C++ in-memory engine, it demonstrates dual-channel real SMS and real email dispatch with microsecond TTL auto-purge and cryptographic single-use DEL eviction. Evaluators can type their own mobile number or email address live during viva and receive their authentic security code on their device in seconds.